EECS 351 Fall 2026 - Project Requirements

1. Project Overview

Propose and implement a data analysis project using digital signal processing (DSP) techniques learned in class, along with at least one technique not covered in class.

2. Technical Requirements

Part 1: Obtain Data
You can use available Datasets. Search for Data online or request data from research groups who work on something that of interest to you. We will also post some resources. Feel Free to collect your own data using your cell phone, computer or audio equipment at the Dude.

Part 2: Apply at Least 3 In-Class DSP Tools

- Processing signals in a different basis
- Frequency domain analysis
- Sampling
- Linear Systems
- Studying system properties
- Low-pass, high-pass, and notch filters
- Image processing techniques

Part 3: Apply at Least 1 Out-of-Class DSP Tool

- Anything that requires processing of data.

3. Project Deliverables

Deliverable 1: Project Website

- Make sure it is easy to navigate
- Make sure you display and explain your topic.
- Discuss your results – strengths and weaknesses of your approach and how you might improve if you were to do it again.
- A page that describes how the three in-class tools were used and how the one out-of-class tool was used.
- A page that says what each team member's contribution was.

Deliverable 2: Code and Data (ZIP File)

- Include a README that describes all dependencies
- Include a small demo – ideally that can run in a short amount of time and show a small scale version of your overall project – as demo.m or demo.py

- If your data is larger than 1GB, and it is publicly accessible, submit only a portion of it, including what you need for the small demo. Put the public link for the data in your README.
- If your data is larger than 1GB and is not publicly accessible, talk to us about your options.

4. Grading Criteria

Projects are evaluated based on:

- Creativity
- Technical Merit and Understanding
- Effort

Across three technical dimensions:

- Mathematics / Theory
- Algorithms / Computational Considerations
- Application

5. Project Timeline

Date	Milestone
Sep 25	Groups finalized
Oct 18	Brainstorm project ideas
Nov 8	Project topic finalized
Nov 15	Progress report
Nov 17	Project peer review
Dec 2–9	Final presentations
Dec 13	Final project submission
Dec 13	Group partner contribution submission